Internet Appendix to

Investors' Attention to Corporate Governance

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January 1, 2021

Abstract

Using unique data on investor views of EDGAR company filings, we document that many investors engage in governance research. However, investors' monitoring is disproportionately focused on large firms and firms with meetings outside the busy spring proxy season. Using an instrumental variables approach that isolates the drop in governance attention during the busy proxy season, we show that governance research is related to investors' monitoring of firms, through voice and exit. Moreover, governance research disciplines management, who, as a result, reduce investments and increase payouts. The concentration of attention results in joint monitoring of a relatively small subset of firms.

Internet Appendix A: EDGAR Data description

Since 1996 the U.S. Securities and Exchange Commission (SEC) has made all company filings publicly available online through the Electronic Data Gathering and Retrieval (EDGAR) database. The EDGAR server records information about each request in the server log files. As discussed by Bauguess, Cooney, and Hanley (2014), these requests exclude two cases. First, they exclude requests of SEC filings from the EDGAR ftp site, which predominantly consist of bulk requests from data vendors. Second, in some cases, ISPs cache frequently requested EDGAR documents for future reference, a scenario that is most likely relevant for the most frequently requested EDGAR documents such as hot IPO prospectuses. The EDGAR server information includes the filing requested, the time and date of the request, and IP addresses of the computer that requested the filing. The SEC has made the server log files created since 2003 available to the public. To ensure a quality match between the investors and their IP addresses, we use these data after 2011.

To protect the privacy of the individuals requesting the filings, the SEC partially masks the IP address that requested the filings. In this paper, we exploit the fact that organizations such as mutual fund families register large blocks of IP address to map individual, partially masked IP address to mutual fund families.

An IP address is composed of four blocks of numbers (octets), each of which ranges from 0 to 255, and this address uniquely identifies the computer. To mask the identity of the IP address of the computer requesting a given filing, the SEC replaces the fourth octet of the IP address with three letters. For example, the IP address 192.175.172.111 might be reported as 192.175.172.dgd in the server logs available for request.

We use a lookup table provided by Digital Elements, a company specializing in analytics and geolocation of IP addresses, to identify the organization(s) that are associated with the first three blocks of each IP address. Continuing with the previous example, the partially masked IP address 192.175.172.dgd likely belongs to Vanguard because "The Vanguard Group, Inc." is the registered owner of all IP addresses that begin with 192.175.172 (i.e., the registered owner of 192.175.172.0, 192.175.172.1, 192.175.172.2, …, 192.175.172.254, 192.175.172.255). We refer to the first three octets of an IP address as an IP3 block.

To form our sample, we start with a list of fund families that have more than 100 voting records in the ISS Voting Analytics database in 2015. For each of these fund families, we use broad regular expressions to match the name with the organizations in the lookup table. For example, for Fidelity Investments we constructed the regular expression (.*fidelity.*)|(.*fmr.*). We manually verify each potential match to create a linking table between fund families and the IP3 blocks of which they are the registered owner.

Most organizations in our sample hold 100% of an IP3 address block, as was the case for Vanguard with the IP3 block 192.175.172. Out of the 47,133 IP3 blocks that we associate with fund families, 95.5% are 100% owned by a single fund family. The remaining 4.5% of IP3 blocks represent cases where an organization in our sample owns a portion of the block. In the cases where one fund

family owns a portion of an IP3 block and the remaining portion of the block is not matched to any fund family in our sample, we assign all EDGAR server activity from that IP3 block to the fund family. In the 110 (0.23%) cases where two or more fund families are registered owners of a fraction of an IP3 block we assign the EDGAR server activity to the fund family that holds the highest fraction of that IP3 block. Finally, there are 71 (0.15%) cases where two or more fund families are registered owners of the same fraction of an IP3 block, and we drop these observations.

The EDGAR server logs record all activity by a user. There are several categories of activity that we exclude. First, the recorded activity includes requests of landing pages, which represent lists of the filings that are available for the user to examine. These requests have an extension of "-index.htm". We view these as uninformative measures of research and exclude them from our analysis of the server logs. Second, we follow Drake, Roulstone, and Thornock (2015) and also exclude clicks on icons (".ico" extensions), XML filings (".xml" extensions), and filings that are under 500 bytes in size. Third, Loughran and McDonald (2017) document that a considerable portion of EDGAR requests are by "robots", which mass request filings for processing through computer programs. We focus on "human" governance research and remove server activity associated with robots. We classify an IP address (e.g., 192.175.172.dgd) as being a robot on a particular day if that IP address requests more than 1,000 filings in a single day. This results in the exclusion of 2,386 robot-day observations by 112 IP addresses that account for 140,853,527 requests. In Table A5, we show that our results are qualitatively similar if we include robot observations. We only exclude the EDGAR server activity from these IP address for the days that IP address is classified as a robot. Finally, if a single IP address requests the same filing multiple times within 5 minutes, we count this as one view of the filing.

The lookup table provided by Digital Elements is a snapshot of all IP address registrations as of April 2016. It is possible that a fund family changes its underlying technology infrastructure and, in that process, changes its registration of IP3 blocks. To minimize the possibility of misattributing EDGAR activity (or lack of EDGAR activity) to a fund family, we use the following methodology to determine a window for which we are confident that the link is high quality. First, for each quarter we calculate the percentage of the fund family's stock holdings on which it conducts research via EDGAR. Starting with the fourth quarter of 2015 we work in reverse chronological order and classify a quarter as a good link if the fund family looks at more than 1% of its positions. We classify the link as no longer valid if two consecutive quarters are below the 1% threshold. To fix ideas, if a fund family uses EDGAR to research 5% in Q4, 22% in Q3, 0% in Q2, and 0% in Q1 in the link would be considered terminated as of the start of the third quarter. Finally, to further avoid the possibility of mismatching the IP addresses, we include EDGAR views data going back only to 2011.

We compare views by these 87 designated mutual fund families, with views by small investors. Small investors are defined as IP addresses that access the internet through common Internet Service Providers (ISPs) and who download a maximum of 25 SEC filings during the calendar year. This yields the following set of ISPs, the root names of which are listed below (i.e., we exclude 'inc', 'llc', etc).

- AT&T
- Bright House Networks

- Charter Communications
- Clear Wireless
- Comcast
- Cox Communications
- Earthlink
- Hughes Network Systems
- Qwest Communications Company
- Sprint
- Time Warner Cable Internet
- T-Mobile USA
- Verizon
- Viasat
- Windstream Communications

Table A1: Determinants of governance research, using broader proxy-related views measure of research

The sample consists of 15,113 firm meetings between 2011 and 2017. In Panel A the dependent variable is governance related research, defined as views of the company's current and prior year proxy statements and any other filings viewed on the same day, from 30 days prior to the release of the current year's proxy statement through the annual meeting. Aggregate views include views by all IP addresses. Mutual fund research includes views by 87 mutual fund families, conditional on them owning shares in the firm. Small investor research includes views by IP addresses that use one of the primary retail Internet Service Providers (e.g., Comcast, Verizon, etc., as defined more fully in Internet Appendix A) and accessed fewer than 25 EDGAR filings during the year. Industry and year fixed effects are included. Variable descriptions are provided in Appendix A. Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1, 5, and 10% levels respectively. Panel B reports variance decompositions. Each entry in the panel represents the Type III partial sum of squares for that independent variable (or set of independent variables in the case of control variables and fixed effects), normalized by the sum across all effects. Within each column, the sum of effects equals one. ***, **, and * denote significance at the 1, 5, and 10% levels respectively.

Panel A: OLS regressions

	Dep't Var = Proxy-related views		
Research by:	Aggregate	Mutual Fund	Small Inv's
Firm Performance			
Market adjusted return	-0.0679***	-0.0858***	-0.0553***
	(0.009)	(0.018)	(0.017)
Profitability	-0.2718***	-0.2968***	-0.3818***
	(0.043)	(0.069)	(0.071)
High default risk	0.2539***	0.3403***	0.4009***
	(0.024)	(0.048)	(0.044)
Firm Size & Financial Characteristics			
Ln(Mkt value of equity)	0.2161***	0.3163***	0.2250***
	(0.003)	(0.006)	(0.005)
Book leverage	0.0741***	0.4075***	-0.0060
	(0.023)	(0.049)	(0.044)
R&D/assets	-0.1188*	0.0091	0.0709
	(0.062)	(0.122)	(0.108)
Cash/assets	-0.0037***	-0.0018	-0.0032**
	(0.001)	(0.002)	(0.001)
Market to book	-0.0017***	-0.0029***	-0.0001
	(0.000)	(0.001)	(0.001)
Tangibility	-0.0227	-0.2449***	-0.0523
	(0.024)	(0.049)	(0.047)
Timing of Firm Meeting			
Busy week of meetings	-0.2570***	-0.1895***	-0.1130***
	(0.008)	(0.017)	(0.016)
Governance Factors			
# of shareholder proposals	0.1337***	0.1033***	0.1988***
	(0.007)	(0.013)	(0.013)
ISS rec. against	-0.0100	0.2875***	0.1304***
	(0.023)	(0.046)	(0.042)
Exempt solicitation	0.1559***	0.1907***	0.2117***
	(0.020)	(0.035)	(0.036)
Revised proxy indicator	0.2436***	0.2177***	0.1614***
	(0.008)	(0.017)	(0.015)
Proxy contest filing	0.3130***	0.3227***	0.2776***
	(0.020)	(0.039)	(0.034)
Has 13D form	0.1429***	0.2800***	0.2159***
	(0.011)	(0.022)	(0.019)
Vote on merger (DEFM14A)	0.1260***	0.1944***	0.1711***
	(0.015)	(0.033)	(0.028)
Fixed effects	Industry, year	Industry, year	Industry, year

Adj. R-squared	0.641	0.364	0.313
Observations	15,113	15,113	15,113

Panel B: Variance Decomposition

	Dep't Var = Proxy-related views		
	Aggregate	Mut Fund	Small Inv's
Market adjusted return	0.45%	0.51%	0.31%
Profitability	0.58%	0.49%	1.19%
High default risk	0.91%	1.15%	2.34%
Fin'l perf. total:	**1.93%**	**2.15%**	**3.84%**
Ln(Mkt value of equity)	43.00%	65.36%	48.39%
Book leverage	0.07%	1.60%	0.00%
R&D/assets	0.03%	0.00%	0.01%
Cash/assets	0.12%	0.02%	0.09%
Market to book	0.12%	0.23%	0.00%
Tangibility	0.01%	0.53%	0.04%
Firm Size & Fin'ls total:	**43.36%**	**67.74%**	**48.53%**
Busy week of meetings	6.97%	2.69%	1.40%
Meeting time total:	**6.97%**	**2.69%**	**1.40%**
# of shareholder proposals	2.95%	1.25%	6.78%
ISS rec. against	0.00%	0.92%	0.28%
Exempt solicitation	0.47%	0.50%	0.90%
Revised proxy indicator	6.50%	3.68%	2.96%
Proxy contest filing	2.10%	1.59%	1.72%
Has 13D form	1.47%	4.01%	3.49%
Vote on merger (DEFM14A)	0.46%	0.78%	0.88%
Gov'ce Total:	**13.95%**	**12.72%**	**17.00%**
Year FE	29.47%	6.67%	7.67%
Industry FE	4.32%	8.03%	21.57%
All	100.00%	100.00%	100.00%

Table A2: Determinants of financial attention and Bloomberg attention over alternative windows

Columns 1 – 3 show variants of regressions in columns 4 and 5 of Table 2, respectively. In column 1, the dependent variable represents views of 10Ks and 10Qs across all IPs, measured over the 60 days following the release of the firm's annual report (instead of over the approximately 80 days preceding the annual meeting). In column 2, the dependent variable is EDGAR views by the 87 mutual funds in our sample, over the same window. In column 3, the dependent variable is the number of mutual fund families (across the 87 in our sample) that view the firm's proxy, over the same window. Column 4 shows a variant of the regression in column 6 of Table 2. The dependent variable is defined as the number of days of Intense Bloomberg search for the company over the ten days following the release of the company's proxy statement (instead of the 10 days prior to the annual meeting). Robust standard errors are reported in parentheses. Panel B shows a variance decomposition analysis, based on these regressions. ***, **, and * denote significance at the 1, 5, and 10% levels respectively.

Panel A: OLS regressions

Type of Research:	#Views of 10K/10Q, over 60 days after filing of 10K	#Views of 10K/10Q, over 60 days after filing of 10K	#MFs viewing 10K/10Q, over 60 days after filing of 10K	Bloomberg, over 10 days following release of proxy statement
Research by:	Aggregate	Mutual Fund	Mutual Fund	Aggregate
Firm Performance				
Market adjusted return	-0.0521***	-0.1069***	-0.0506***	-0.0068
	(0.009)	(0.018)	(0.010)	(0.007)
Profitability	-0.1927***	-0.1304**	-0.0265	-0.0829***
	(0.036)	(0.065)	(0.037)	(0.030)
High default risk	0.2133***	0.4212***	0.2527***	0.0682***
	(0.024)	(0.050)	(0.027)	(0.023)
Firm Size &Financial Characteristics				
Ln(Mkt value of equity)	0.1932***	0.5229***	0.2820***	0.0399***
	(0.004)	(0.006)	(0.003)	(0.002)
Book leverage	0.0637**	0.9903***	0.5409***	0.0061
	(0.027)	(0.050)	(0.026)	(0.018)
R&D/assets	0.0086	-0.0236	0.1613**	0.0093
	(0.078)	(0.125)	(0.069)	(0.047)
Cash/assets	-0.0043***	0.0018	0.0007	0.0009
	(0.001)	(0.002)	(0.001)	(0.001)
Market to book	-0.0008	-0.0034***	-0.0016***	-0.0003
	(0.001)	(0.001)	(0.000)	(0.000)
Tangibility	0.1053***	0.0730	0.0166	0.0226
	(0.030)	(0.052)	(0.028)	(0.018)
Timing of Firm Meeting				
Busy week of meetings	-0.1715***	-0.0505***	-0.0326***	0.0075
	(0.010)	(0.017)	(0.009)	(0.006)
Governance Factors				
# of shareholder proposals	0.1234***	-0.0127	-0.0202***	0.0051
	(0.011)	(0.016)	(0.008)	(0.006)
ISS rec. against	-0.0362	-0.0647	-0.0284	-0.0181
	(0.024)	(0.048)	(0.025)	(0.015)
Exempt solicitation	0.1356***	0.0751*	0.0483**	0.0041
	(0.029)	(0.044)	(0.023)	(0.017)
Revised proxy indicator	0.0168*	0.0782***	0.0553***	0.0034
	(0.009)	(0.017)	(0.009)	(0.006)
Proxy contest filing	0.1348***	0.1631***	0.0671***	0.0154
	(0.024)	(0.040)	(0.021)	(0.016)
Has 13D form	0.0186*	0.1488***	0.0788***	0.0224***
	(0.011)	(0.022)	(0.011)	(0.008)
Vote on merger (DEFM14A)	0.0600***	-0.0323	-0.0125	-0.0029
	(0.018)	(0.034)	(0.019)	(0.014)
Fixed effects	Industry, year	Industry, year	Industry, year	Industry, year
Adj. R-squared	0.707	0.587	0.615	0.073
Observations	12,250	12,250	12,250	10,547

Panel B: Variance decomposition

	#Views of 10K, over 60 days after filing of 10K	#Views of 10K, over 60 days after filing of 10K	#MFs viewing 10K, over 60 days after filing of 10K	Bloomberg, over 10 days following release of proxy statement
	Aggregate	Mutual Fund	Mutual Fund	Aggregate
Firm performance	0.67%	1.13%	1.13%	4.16%
Firm financial characteristics	16.46%	69.11%	66.32%	58.20%
Timing of firm meeting	1.53%	0.07%	0.09%	0.24%
Governance factors	2.55%	0.85%	0.91%	2.15%
Industry fixed effects	6.92%	17.94%	20.22%	20.82%
Year fixed effects	71.86%	10.91%	11.33%	14.43%

Table A3: Governance research of passive versus active investors across different firms

This table shows regressions similar to those reported in column 1 of Table 4, where the sample consists of an unbalanced panel of 87 mutual fund families×firms owned by each fund, between 2011 and 2017. The difference is that here we include interactions between whether a mutual fund family includes a large portion of indexed assets ('Indexer') and various measures of the contentiousness of the shareholder meeting. *Indexer* equals one if the fraction of total net assets held by index funds in a firm family is greater than 50 percent, zero otherwise. The measures of contentiousness include: *# of shareholder proposals*, *ISS recommend against*, *Has 13D form*, *Proxy contest filing*, and *Exempt solicitation*, all of which are defined in Appendix A. All control variables from Table 4 are also included (not tabulated), plus industry and year fixed effects. Standard errors are clustered by firm-meeting. ***, **, and * denote significance at the 1, 5, and 10% levels respectively.

	Dep't Variable = Proxy-related views				
Indexer x # Shareholder proposals	0.0107***				
	(0.002)				
Indexer x ISS rec. against		0.1565***			
		(0.010)			
Indexer x Has 13D Form			0.0194***		
			(0.005)		
Indexer x Proxy contest filing				0.0148	
				(0.009)	
Indexer x Exempt solicitation					0.0118*
					(0.007)
# of shareholder proposals	0.0080***	0.0097***	0.0094***	0.0095***	0.0094***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
ISS rec. against	0.0593***	0.0325***	0.0594***	0.0593***	0.0593***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Has 13D form	0.0171***	0.0172***	0.0140***	0.0172***	0.0172***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Proxy contest filing	0.0466***	0.0470***	0.0465***	0.0444***	0.0466***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
Exempt solicitation	0.0090	0.0088	0.0086	0.0086	0.0071
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R-squared	0.087	0.088	0.087	0.087	0.087
Observations	405,543	405,543	405,543	405,543	405,543

Table A4: Where do investors conduct governance-related research? With investor, firm, and firm-meeting fixed effects

This table shows regressions similar to those in Table 4, where the sample consists of an unbalanced panel of 87 mutual fund families×firms owned by each fund, between 2011 and 2017. In columns 1, 3, and 5, for each investor×firm meeting, the dependent variable equals the log of one plus the investor's views of the firm's current and prior year proxy statements. In columns 2, 4, and 6, the dependent variable is an indicator variable equal to one if the investor viewed one or more firm proxies, zero otherwise. Columns 1 and 2 (columns 3 and 4) show regressions similar to those in Columns 1 and 3 of Table 4, with the exception that we include investor fixed effects (firm fixed effects). The specifications in columns 5 and 6 are similar, but include firm-meeting fixed effects. This compares to the use of the industry and year fixed effects used in Table 4. Standard errors are clustered by firm-meeting. ***, **, and * denote significance at the 1, 5, and 10% levels, respectively.

	Proxy views	Proxy views Extensive Margin	Proxy views	Proxy views Extensive Margin	Proxy views	Proxy views Extensive Margin
Investor Characteristics						
Investment as % of firm equity	0.0084***	0.0055***	0.0207***	0.0187***	0.0208***	0.0188***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Investment as % of mutual fund	0.0458***	0.0296***	0.0073*	0.0033	0.0082**	0.0039
family equity position	(0.004)	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)
Ln(Fund family equity holdings)	-0.0062***	-0.0121***	0.0238***	0.0209***	0.0238***	0.0209***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Fraction index funds			-0.0413***	-0.0220***	-0.0418***	-0.0225***
			(0.002)	(0.002)	(0.002)	(0.002)
Firm Performance						
Market adjusted return	-0.0090***	-0.0053***	-0.0046***	-0.0030**		
	(0.002)	(0.001)	(0.001)	(0.001)		
Profitability	-0.0527***	-0.0349***	-0.0173*	-0.0113		
	(0.007)	(0.005)	(0.010)	(0.008)		
High default risk	0.0354***	0.0249***	-0.0007	-0.0013		
	(0.005)	(0.004)	(0.005)	(0.004)		
Firm Size & Financial Characteristics						
Ln(Mkt value of equity)	0.0335***	0.0255***				
	(0.001)	(0.000)				
Book leverage	0.0134***	0.0126***				
	(0.005)	(0.003)				
R&D/assets	-0.0397***	-0.0285***	-0.0271	-0.0210		
	(0.012)	(0.009)	(0.019)	(0.016)		
Cash/assets	-0.0004**	-0.0003	-0.0000	-0.0001		
	(0.000)	(0.000)	(0.000)	(0.000)		
Market to book	-0.0002***	-0.0001**				
	(0.000)	(0.000)				
Tangibility	-0.0204***	-0.0124***				
	(0.004)	(0.003)				

Timing of Firm Meeting						
Busy week of meetings	-0.0189***	-0.0139***				
	(0.002)	(0.001)				
Governance Factors						
# of shareholder proposals	0.0177***	0.0098***	0.0096***	0.0078***		
	(0.002)	(0.001)	(0.002)	(0.001)		
ISS rec. against	0.0410***	0.0273***	0.0594***	0.0388***		
	(0.005)	(0.003)	(0.006)	(0.004)		
Exempt solicitation	0.0211***	0.0127***	0.0093*	0.0056		
	(0.005)	(0.003)	(0.005)	(0.004)		
Revised proxy indicator	0.0194***	0.0084***	0.0189***	0.0091***		
	(0.001)	(0.001)	(0.002)	(0.002)		
Proxy contest filing	0.0359***	0.0205***	0.0467***	0.0260***		
	(0.005)	(0.003)	(0.010)	(0.006)		
Has 13D form	0.0234***	0.0141***	0.0173***	0.0110***		
	(0.002)	(0.002)	(0.002)	(0.002)		
Vote on merger (DEFM14A)	0.0114***	0.0077***	0.0372***	0.0243***		
	(0.004)	(0.002)	(0.008)	(0.005)		
Fixed effects	Investor, industry, year	Investor, industry, year	Firm, year	Firm, year	Meeting	Meeting
Adj. R-squared	0.210	0.232	0.085	0.080	0.082	0.074
Observations	405,543	405,543	405,543	405,543	405,543	405,543

Table A5: Where do investors conduct governance-related research, including views by bots

This table shows regressions similar to those in the analogous columns of Table 4, where the sample consists of an unbalanced panel of 87 mutual fund families×firms owned by each fund, between 2011 and 2017. The only difference is that we include views by bots, where a bot is defined as a single IP address that views 1,000 or more filings within a single day. Standard errors are clustered by firm-meeting. ***, **, and * denote significance at the 1, 5, and 10% levels, respectively.

	Proxy views	Proxy-related Views	Proxy views Extensive Margin	Proxy Views Intensive Margin
Investor Characteristics				
Investment as % of firm equity	0.0218***	0.0269***	0.0188***	0.0071***
	(0.000)	(0.001)	(0.000)	(0.001)
Investment as % of fund family equity holdings	0.0173***	0.0118**	0.0081***	0.0141**
	(0.004)	(0.005)	(0.003)	(0.007)
Ln(Fund family equity holdings)	0.0277***	0.0349***	0.0231***	0.0344***
	(0.000)	(0.000)	(0.000)	(0.001)
Fraction index funds	-0.0711***	-0.1068***	-0.0370***	-0.2709***
	(0.002)	(0.002)	(0.001)	(0.009)
Firm Performance				
Market adjusted returns	-0.0112***	-0.0125***	-0.0068***	-0.0318***
	(0.002)	(0.002)	(0.001)	(0.006)
Profitability	-0.0545***	-0.0694***	-0.0378***	-0.1107***
	(0.007)	(0.009)	(0.005)	(0.025)
High default risk	0.0389***	0.0507***	0.0277***	0.0509***
	(0.005)	(0.007)	(0.004)	(0.015)
Firm Size & Financial Characteristics				
Ln(Mkt value of equity)	0.0270***	0.0329***	0.0193***	0.0525***
	(0.001)	(0.001)	(0.000)	(0.002)
Book leverage	0.0136***	0.0265***	0.0116***	0.0360**
	(0.005)	(0.006)	(0.003)	(0.017)
R&D/assets	-0.0319**	-0.0365**	-0.0238**	-0.0381
	(0.012)	(0.016)	(0.009)	(0.045)
Cash/assets	-0.0004**	-0.0005*	-0.0003	-0.0009
	(0.000)	(0.000)	(0.000)	(0.001)
Market to book	-0.0003***	-0.0003***	-0.0001**	-0.0007**
	(0.000)	(0.000)	(0.000)	(0.000)
Tangibility	-0.0220***	-0.0305***	-0.0126***	-0.0602***
	(0.005)	(0.006)	(0.003)	(0.017)
Timing of Firm Meeting				
Busy week of meetings	-0.0173***	-0.0193***	-0.0125***	-0.0392***
	(0.002)	(0.002)	(0.001)	(0.006)
Governance Factors				
# of shareholder proposals	0.0198***	0.0222***	0.0112***	0.0194***
	(0.002)	(0.002)	(0.001)	(0.004)
ISS rec. against	0.0483***	0.0567***	0.0330***	0.1325***
	(0.005)	(0.006)	(0.003)	(0.017)
Exempt solicitation	0.0213***	0.0262***	0.0133***	0.0256**
	(0.005)	(0.006)	(0.003)	(0.013)
Revised proxy indicator	0.0205***	0.0219***	0.0077***	0.1134***
	(0.001)	(0.002)	(0.001)	(0.006)
Proxy contest filing	0.0388***	0.0453***	0.0209***	0.0962***
	(0.005)	(0.006)	(0.003)	(0.016)
Has 13D form	0.0283***	0.0393***	0.0174***	0.0661***
	(0.003)	(0.003)	(0.002)	(0.008)
Vote on merger (DEFM14A)	0.0139***	0.0209***	0.0089***	0.0270**
	(0.004)	(0.005)	(0.002)	(0.012)
Fixed effects	Industry, year	Industry, year	Industry, year	Industry, year
Adj. R-squared	0.078	0.068	0.077	0.109
Observations	405,543	405,543	405,543	46,606

Table A6: Where do investors conduct governance-related research? A multinomial logit with relative risk parameters

This table shows a multinomial logit regression, where we define four possible scenarios for investor×firm meeting governance research: no research (which represents the omitted category), research prior to (but not after) the estimated release of ISS's recommendation (reported in column 1), research following (but not prior to) the estimated release of ISS's recommendation (reported in column 2), and research by prior to and after the estimated release of ISS's recommendation (reported in column 3). In each case, research is defined as a dummy equal to one if the investor viewed the firm's proxy statement in the window beginning 30 days prior to the release of the current year proxy and ending at the annual meeting, zero otherwise. The estimated release of ISS's recommendation is set equal to 14 days prior to the firm's annual meeting. We report relative risk ratio parameters, which reflect the percentage effect of the independent variable on the outcome, relative to 1. For example, 1.19 should be interpreted as a 19% increased probability, whereas 0.75 should be interpreted as a 25% decreased probability. Standard errors are clustered by firm-meeting. ***, **, and * denote significance at the 1, 5, and 10% levels, respectively.

	Dep't Variable = 1 if investor viewed a firm proxy in the designated window		
Omitted Category: No proxies viewed	Only Pre-ISS Rec	Only Post-ISS Rec	Pre- and Post-ISS Rec
Investor Characteristics			
Investment as % of firm equity	1.0813***	1.1222***	1.0959***
	(0.002)	(0.005)	(0.008)
Investment as % of fund family equity holdings	1.1977***	1.1719***	1.1850***
	(0.026)	(0.066)	(0.064)
Ln(Fund family equity holdings)	1.4803***	1.3476***	2.0194***
	(0.007)	(0.017)	(0.054)
Fraction index funds	0.5653***	0.9609	0.0832***
	(0.011)	(0.055)	(0.011)
Firm Performance			
Market adjusted returns	0.9237***	0.8816*	0.6960***
	(0.015)	(0.058)	(0.068)
Profitability	0.7062***	0.6257***	0.4413***
	(0.042)	(0.102)	(0.085)
High default risk	1.2349***	1.4059***	1.6561***
	(0.050)	(0.158)	(0.228)
Firm Size & Financial Characteristics			
Ln(Mkt value of equity)	1.2356***	1.2581***	1.6939***
	(0.006)	(0.021)	(0.039)
Book leverage	1.1718***	1.1067	1.4021*
	(0.045)	(0.151)	(0.254)
R&D/assets	0.8244*	0.5366*	0.3341**
	(0.092)	(0.189)	(0.179)
Cash/assets	0.9985	0.9812**	0.9785
	(0.002)	(0.008)	(0.015)
Market to book	0.9987**	0.9981	0.9988
	(0.001)	(0.002)	(0.002)
Tangibility	0.8894***	0.8783	0.6852**
	(0.035)	(0.115)	(0.119)
Timing of Firm Meeting			
Busy week of meetings	0.8812***	0.6731***	0.6872***
	(0.012)	(0.034)	(0.044)
Governance Factors			
# of shareholder proposals	1.0601***	1.0129	1.0704***
	(0.010)	(0.028)	(0.025)
ISS rec. against	1.2860***	2.8638***	1.9997***
	(0.049)	(0.313)	(0.345)
Exempt solicitation	1.0797***	1.1582*	1.1901**
	(0.029)	(0.093)	(0.102)
Revised proxy indicator	1.0852***	1.2418***	1.4042***

	(0.014)	(0.062)	(0.098)
Proxy contest filing	1.2404***	0.9192	1.3920***
	(0.036)	(0.075)	(0.150)
Has 13D form	1.1645***	1.1100*	1.2792***
	(0.020)	(0.063)	(0.093)
Vote on merger (DEFM14A)	1.0721***	1.0401	1.1885**
	(0.026)	(0.076)	(0.100)
Fixed effects		Industry, year	
Pseudo R-squared		0.135	
Observations		405,543	
Chi-squared test		982,7623	

Table A7: Coordination using a continuous measure

This table presents regressions similar to those in the analogous columns of Table 5, where the sample consists of an unbalanced panel of 87 mutual fund families×firms owned by each fund, between 2011 and 2017. The only difference is that here we use continuous measures of research. For each investor×firm meeting, the dependent variable is the log of one plus the investor's views of the firm's current and prior year proxy statements in the window beginning 30 days prior to the release of the current year proxy statement and continuing through the annual meeting. To calculate '*Other investors' expected governance research*' we first define expected research for each fund family, equal to the log of one plus views of proxy statements in the preceding year before the firm meetings. Second, for each fund family×firm meeting observation, we calculate the ownership-weighted expected governance research of all other fund families, where weights are based on ownership for the current year, i.e., the quarter-end prior to the meeting. Regressions include industry, year, and investor fixed effects, as well as all controls included in Table 4 (not tabulated). Standard errors are clustered by firm-meeting. ***, **, and * denote significance at the 1, 5, and 10% levels respectively.

	Dep't Variable = Proxy Views				
Other investors' expected governance research	0.0181	0.0081	0.0166	0.0229**	0.0313***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.012)
Other investors' expected governance research × High investment as % of firm equity		0.1104***			
		(0.021)			
Other investors' expected governance research × High investment as % of fund familiy equity holdings			0.0784***		
			(0.015)		
Other investors' expected governance research × Market adj. return				-0.0872***	
				(0.020)	
Other investors' expected governance research × Profitability					-0.1407***
					(0.048)
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	Industry, year, Investor	Industry, year, Investor	Industry, year, Investor	Industry, year, Investor	Industry, year, Investor
Adj. R-squared	0.232	0.232	0.232	0.232	0.232
Observations	397,165	397,165	397,165	397,165	397,165

Table A8: Comparison of firm-meetings in busy versus non-busy periods

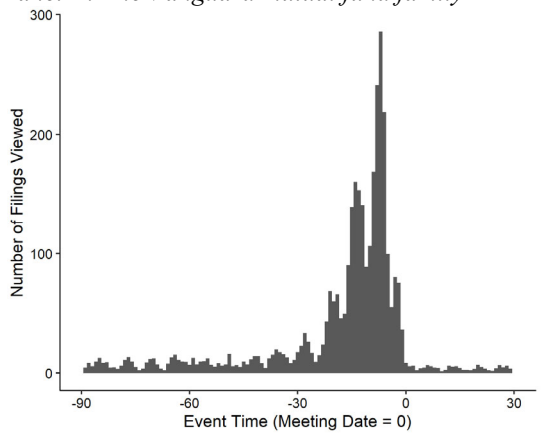
This table compares the characteristics of firm meetings that fall into the busy Spring proxy season, defined as weeks 18 – 24 during each calendar year, with meetings that fall into all other weeks of the calendar year. We present univariate comparisons of each of our measures of meeting contentiousness.

	Busy weeks (N = 10,648)	Non-Busy weeks (N = 4,465)
Proxy contest filing	0.045	0.042
Merger filing	0.063	0.060
Has 13D form	0.167	0.164
Number of shareholder proposals	0.227	0.128
ISS recommend against	0.099	0.102
Exempt solicitation	0.059	0.040
Revised proxy indicator	0.539	0.520

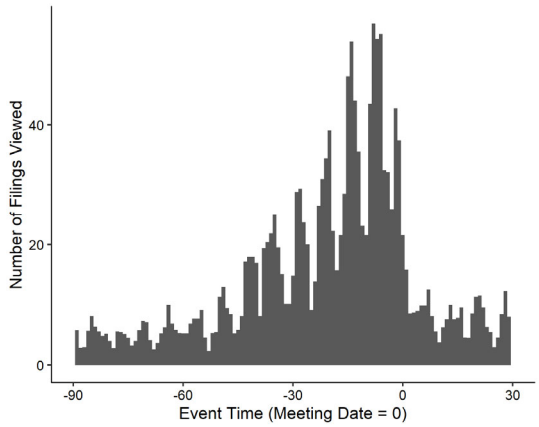
Figure A1: Total governance-related research of three mutual fund families

The sample in Panel A consists of event-time governance-related filings viewed by the Vanguard mutual fund family for firms in their portfolio, between 2011 and 2017. For each firm-year, we focus on Vanguard's views of the firm's proxy statements, including the current year and prior years. In the figure, we count the number of views of these filings from calendar days -90 until day 30, where the company's annual meeting represents day 0. Panel B (Panel C) is similar to Panel A but represents views by the Fidelity (Blackrock) mutual fund family.

Panel A: The Vanguard mutual fund family



Panel B: The Fidelity mutual fund family



Panel C: The Blackrock mutual fund family

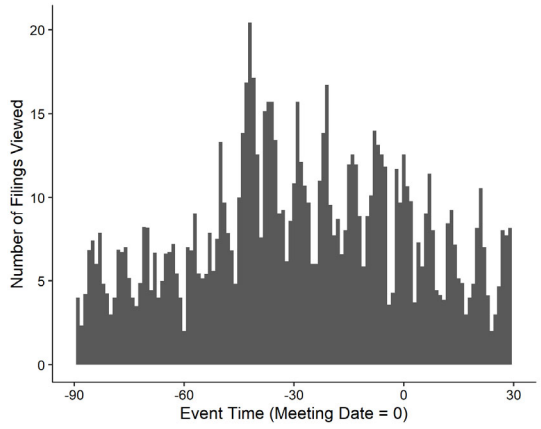


Figure A2: Governance-related research by mutual fund families in publicly traded firms

The sample consists of an unbalanced panel of firms held by 87 mutual fund families, between 2011 and 2017. For each investor×firm meeting, we focus on the investor's views of the firm's proxy current and prior year proxy statements. We count the number of views of these filings, from 30 days prior to the release of the proxy statement through the date of the annual meeting, and the figures show the day of the week of the views.

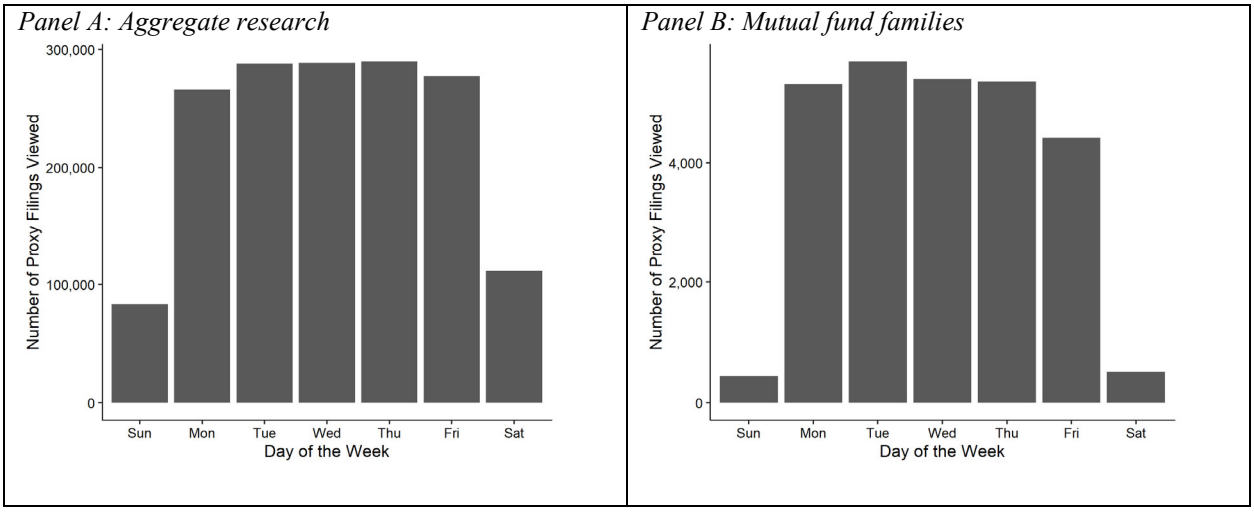


Figure A3: Timing of proxy release date, relative to the company meeting

This figure shows the number of days prior to the annual meeting that the proxy statement is posted to EDGAR for all firms in the sample.

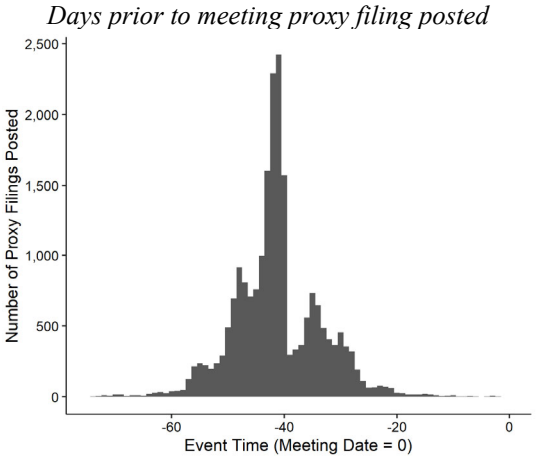
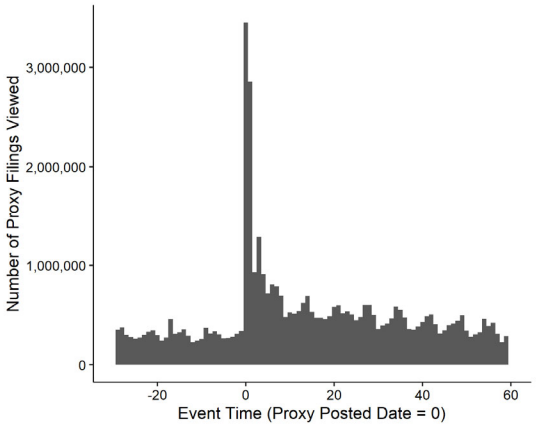


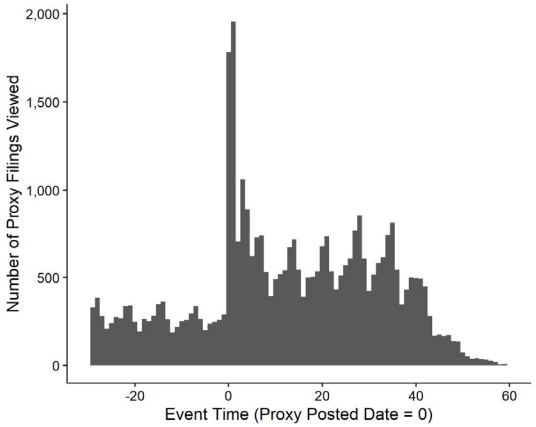
Figure A4: Total proxy filing reads relative to the proxy filing posted date

The sample consists of all EDGAR users (Panel A), an unbalanced panel of firms held by 87 mutual fund families (Panel B), or all small investors (Panel C) between 2011 and 2017. Each panel presents the number of views of any proxy statement in event time, where the event date is defined as the date the current year's proxy statement is posted to EDGAR. We focus on the window beginning 30 days prior to the release of the proxy statement and continuing until 60 days after.

Panel A: Views of any proxy in aggregate



Panel B: Views of any proxy by mutual funds



Panel C: Views of any proxy by small investors

